

The bot is failing because it isn't doing math, instead it is predicting what the "most probable" output of that equation will be. As we discussed towards the end of class, the prediction works for common problems found within training sets (basic arithmetic, algebra, even calculus); but the prediction fails when it gets to a point where the operation is no longer common enough to be a probable answer. A good example of this I found was when multiplying 10's ($n=10$ $i=10$); everything up to 6 it did fine, but for every equation it wouldn't follow any structure for generating zeros, it just put what was the 'most likely' output; with the invalidity of its output becoming consistent after iteration 6. If the equation it is predicting isn't commonly stated in the sources scraped for training data, the answer is most likely going to be wrong, as there are no explicit examples to base the answer off.

Transformer architecture doesn't and will never actually calculate anything (barring further development); all it does is predict based on information within the context window on what (roughly speaking) a correct answer would look like. Even if it does provide a 'correct answer' to the function(n,i) , no calculations are ever actually being done. Meaning if it doesn't fail, then the calculation(s) are common enough to give sufficient training data for the transformer architecture's most probable answer to also be the correct one.